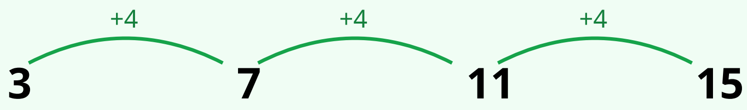
Quadratic Sequences

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



nth term = 4n - 1

Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

- Forgetting to halve the second difference.
- Slips subtracting the an^2 values.
- Not finding the linear part correctly.
- Sign errors with a negative linear part.

METHOD

1. First differences
2. Second differences (constant?)
3. a = second difference ÷ 2
4. Subtract an^2 find $bn + c$
5. Combine: $an^2 + bn + c$

WORKED EXAMPLE

Find nth term of 3, 8, 15, 24, 35.
Second difference 2 a = 1 (n^2). Subtract n^2: 2, 4, 6, 8, 10 = 2n.
nth term = $n^2 + 2n$.

VISUAL AID

A quadratic sequence has a constant SECOND difference. Halving it gives the coefficient of n^2; subtracting the an^2 values leaves a simpler linear sequence to finish the rule.

1. (2 marks)
Find first and second differences: a) 2, 5, 10, 17, 26 b) 4, 7, 12, 19, 28

2. (2 marks)
Quadratic? Show second differences: a) 1, 4, 9, 16 b) 2, 4, 6, 8

3. (3 marks)
Find the nth term: a) 2, 5, 10, 17, 26 b) 4, 7, 12, 19, 28 c) 3, 9, 19, 33, 51

4. (3 marks)
nth term = $n^2 + n$. a) 1st term b) 5th term c) 10th term

5. (3 marks)
Find the nth term: a) 0, 3, 8, 15, 24 b) 5, 14, 27, 44, 65 c) 3, 8, 15, 24

6. (3 marks)
Tile pattern: 2, 6, 12, 20, ... a) Tiles in pattern 5. b) nth term.

7. (4 marks)
Sequence 1, 6, 15, 28, 45. a) Show second difference constant. b) nth term. c) 10th term.

8. (5 marks)
nth term = $n^2 - 3n + 2$. a) First four terms. b) Show 2nd difference is 2. c) Which term = 30?